

October 28, 2025

Retrieval-Augmented Generation

Outline

Background

Challenges & Possible improvement ideas

Hands-on with RAG

References:

UCB EECS CS 288. Natural Language Processing

Sharma, C. "Retrieval-Augmented Generation: A Comprehensive Survey of Architectures, Enhancements, and Robustness Frontiers." in arXiv preprint arXiv:2506.00054, 2025.

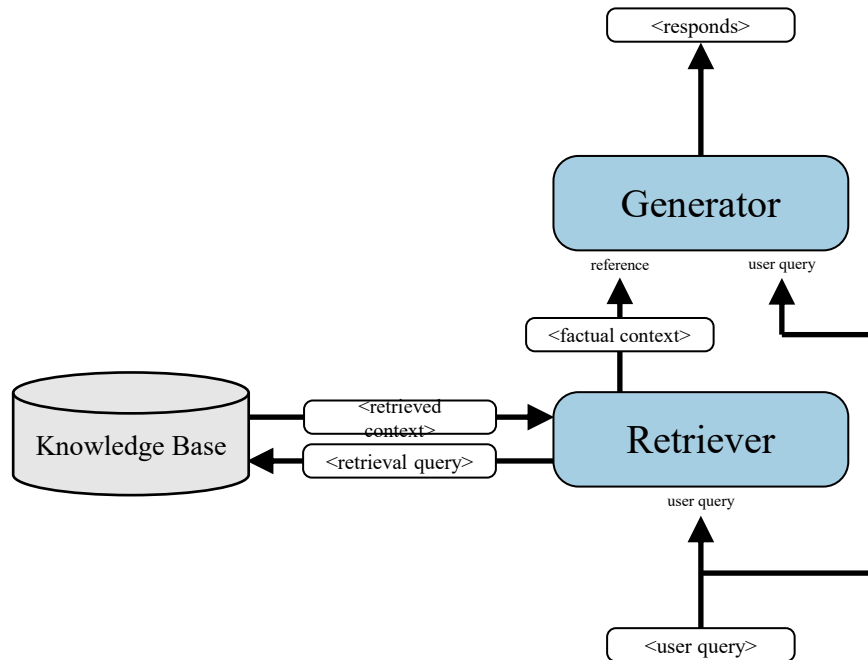
Outline

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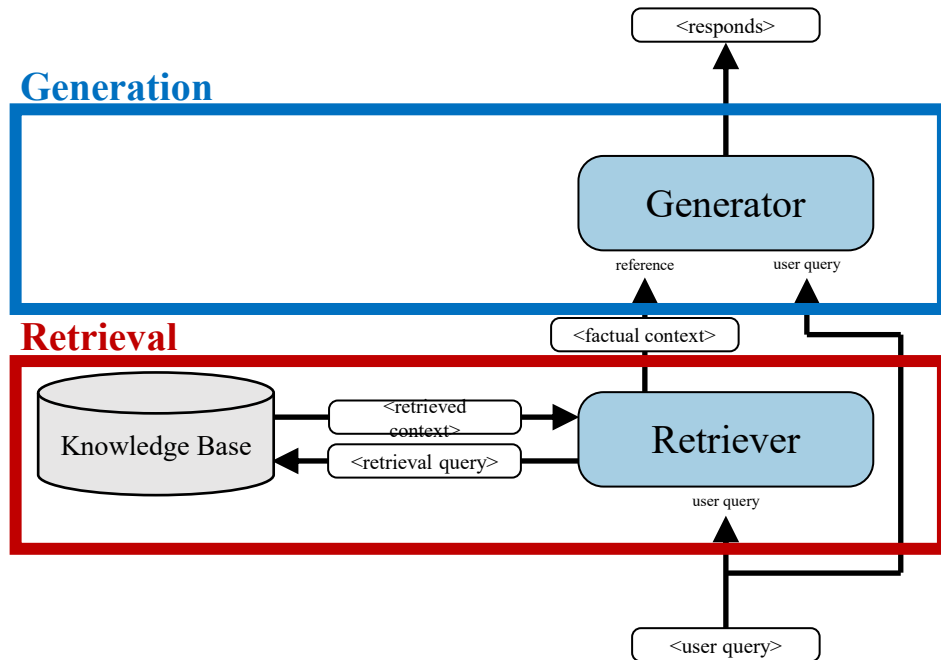
Challenges & Possible improvement ideas

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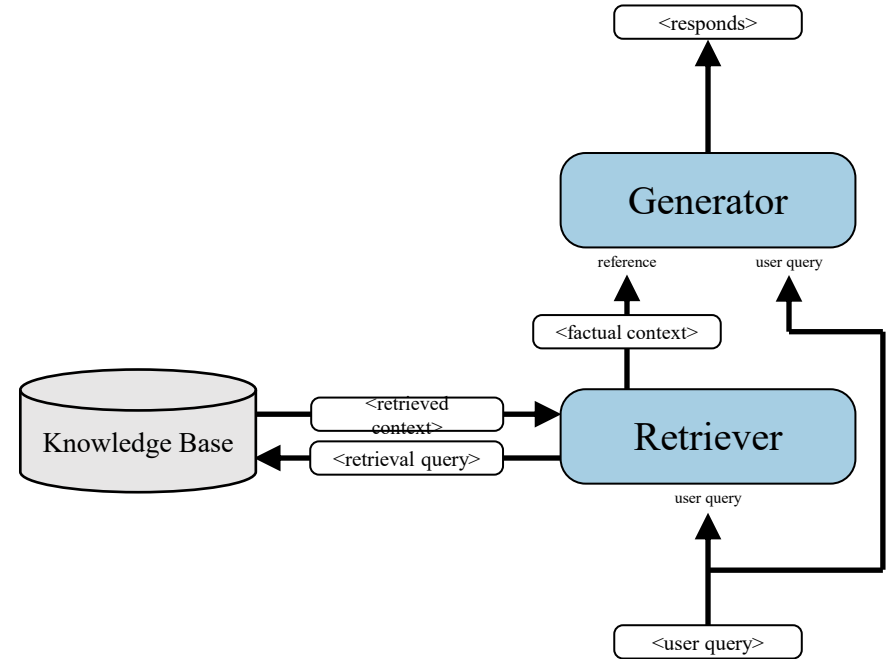
Challenges & Possible improvement ideas

Challenges & Possible improvement ideas – (Question Answering in specific domain)

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Challenges

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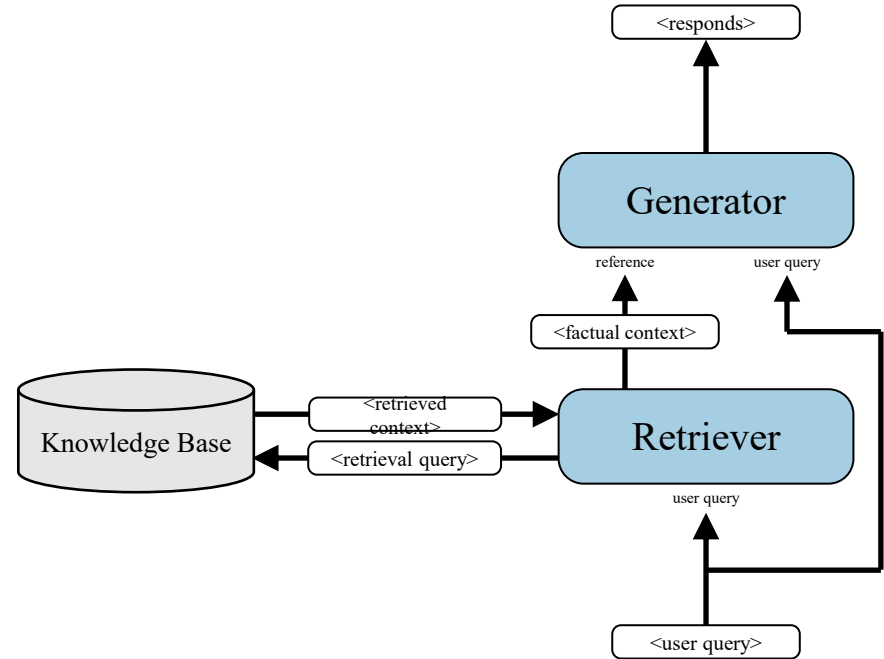


Challenges & Possible improvement ideas – (Question Answering in specific domain)

Challenges

- Retrieval quality

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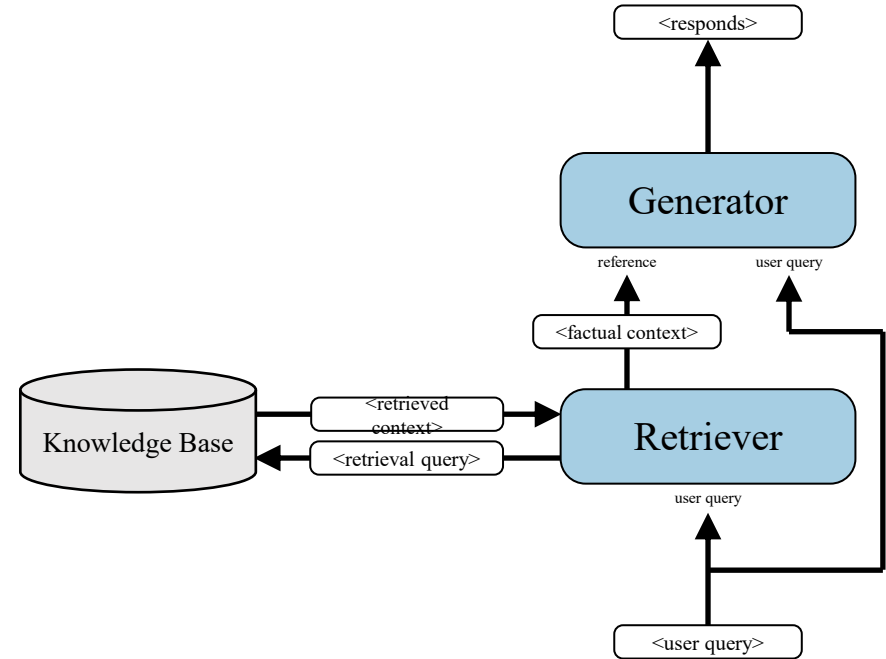


Challenges & Possible improvement ideas – (Question Answering in specific domain)

Challenges

- Retrieval quality, Grounding fidelity

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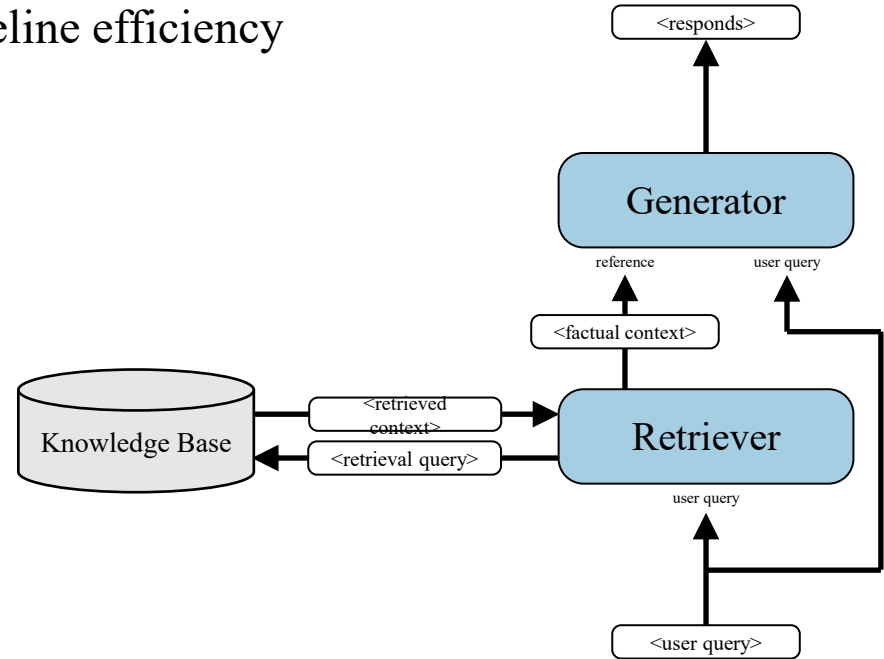


Challenges & Possible improvement ideas – (Question Answering in specific domain)

Challenges

- Retrieval quality, Grounding fidelity, Pipeline efficiency

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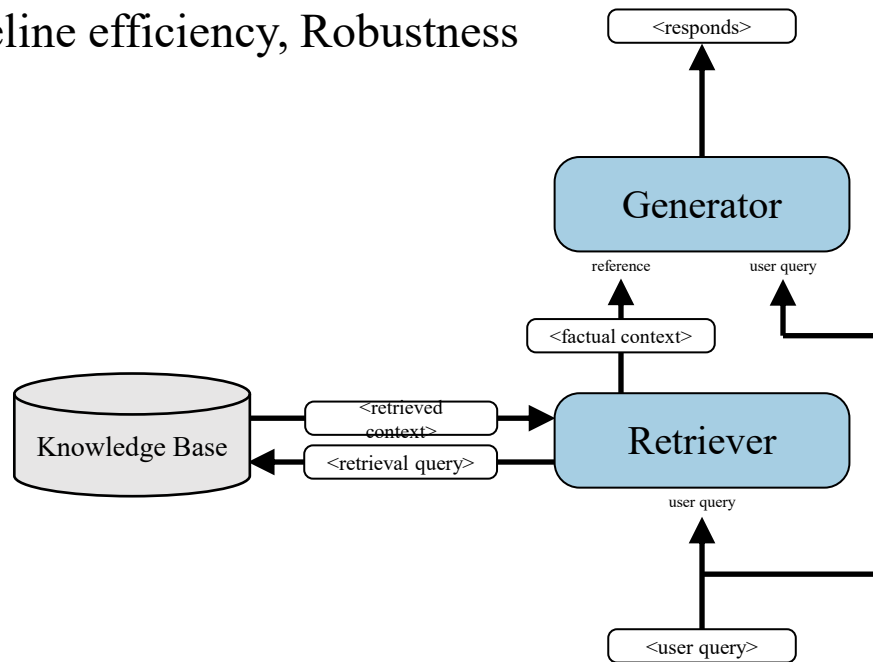


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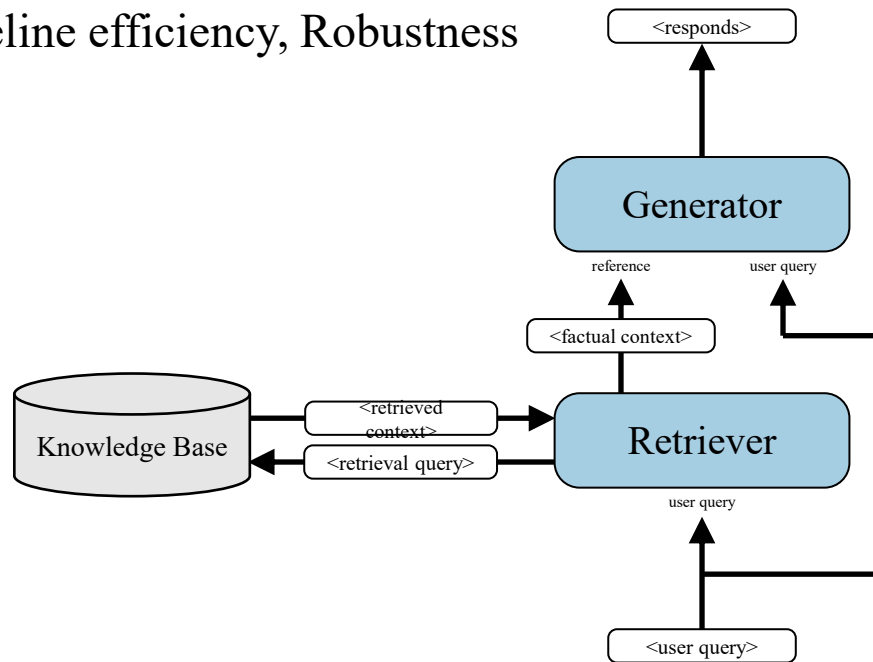
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Possible improvement ideas

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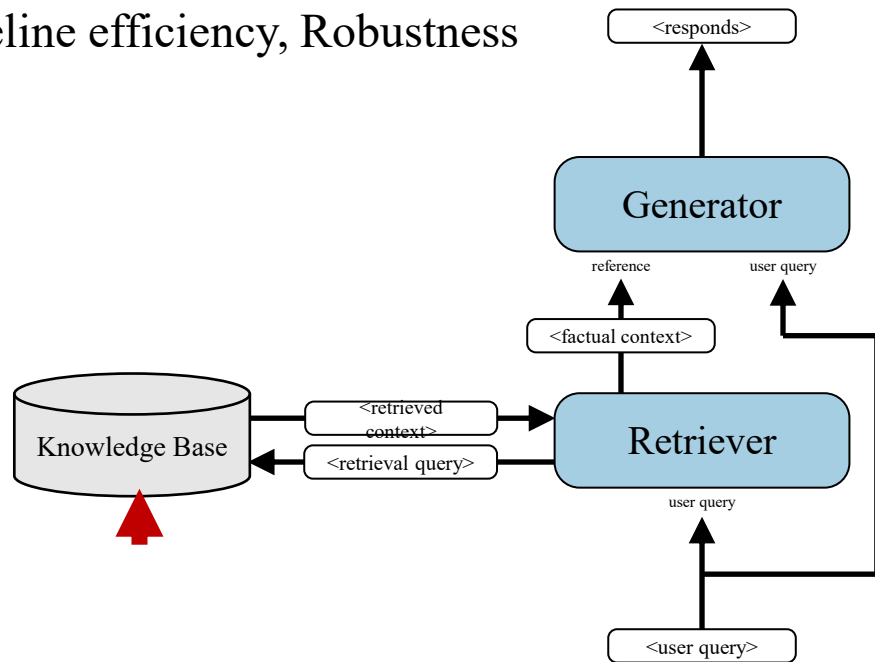
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Possible improvement ideas

- Chunking

What should be the retrieval unit? Document? Passage?

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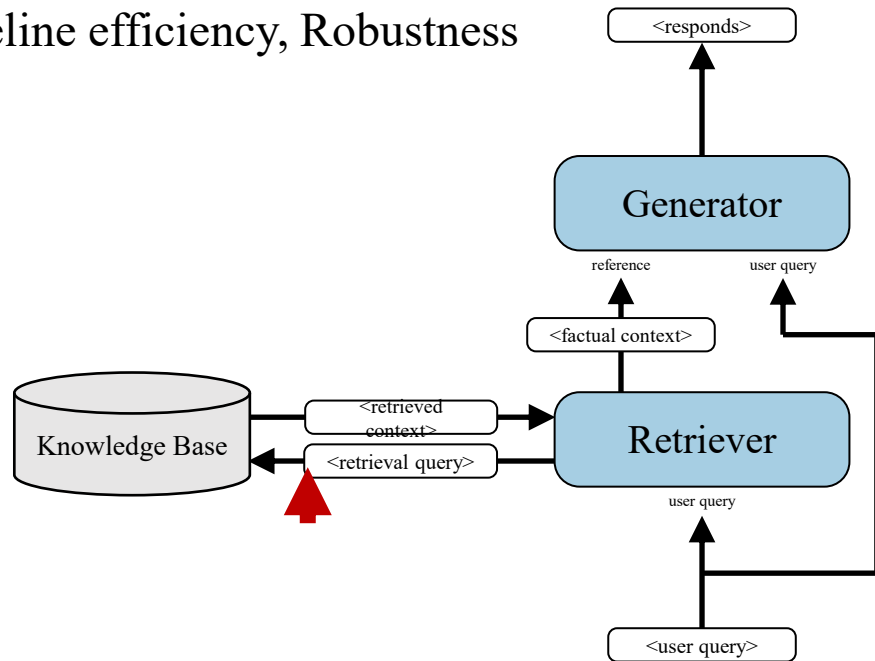
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- Retrieval method

Sparse retrieval (BM25) or dense retrieval? Or hybrid?

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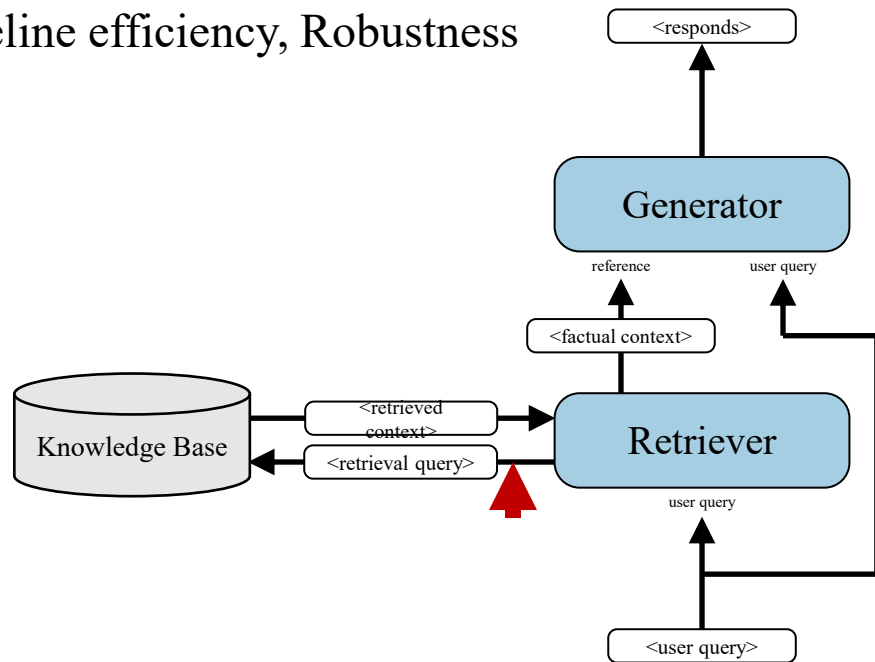
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Is user input suitable for direct use as a query?



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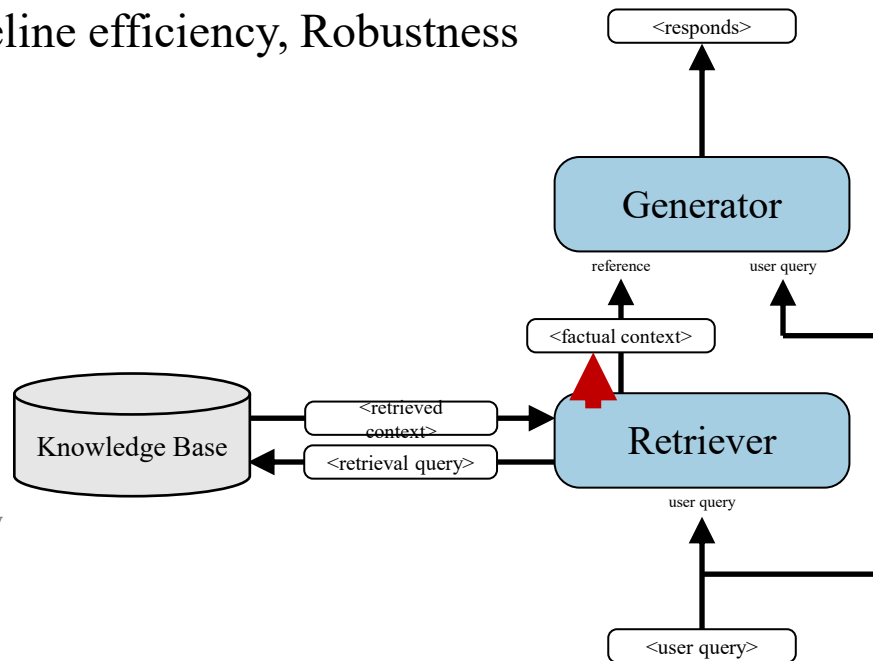
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Is user input suitable for direct use as a query?

- Rewriting

make passages more comprehensible & include relevant info only

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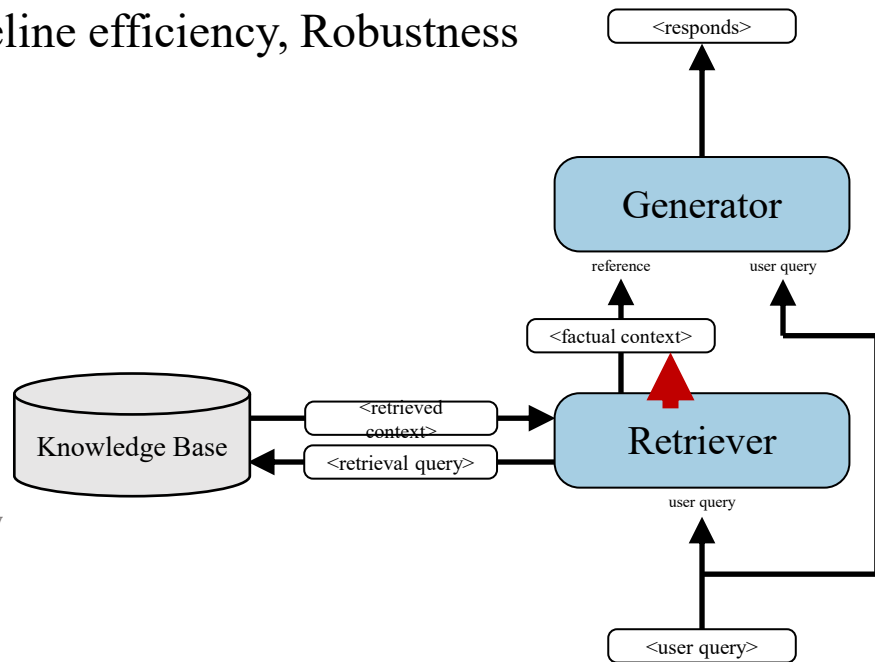
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make passages more comprehensible & include relevant info only
- Reranking/adaptive adoption
use only relevant retrieved passages
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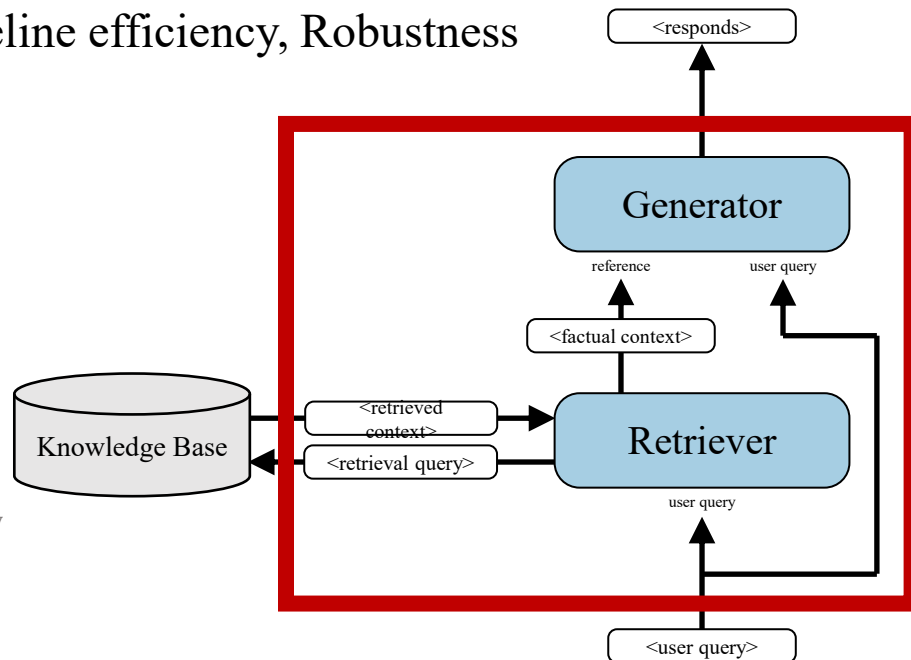
make passages more comprehensible & include relevant info only

- Reranking/adaptive adoption

use only relevant retrieved passages

- Pipeline & Framework design

Deciding when to use retrieval? Multiple rounds of retrieval?



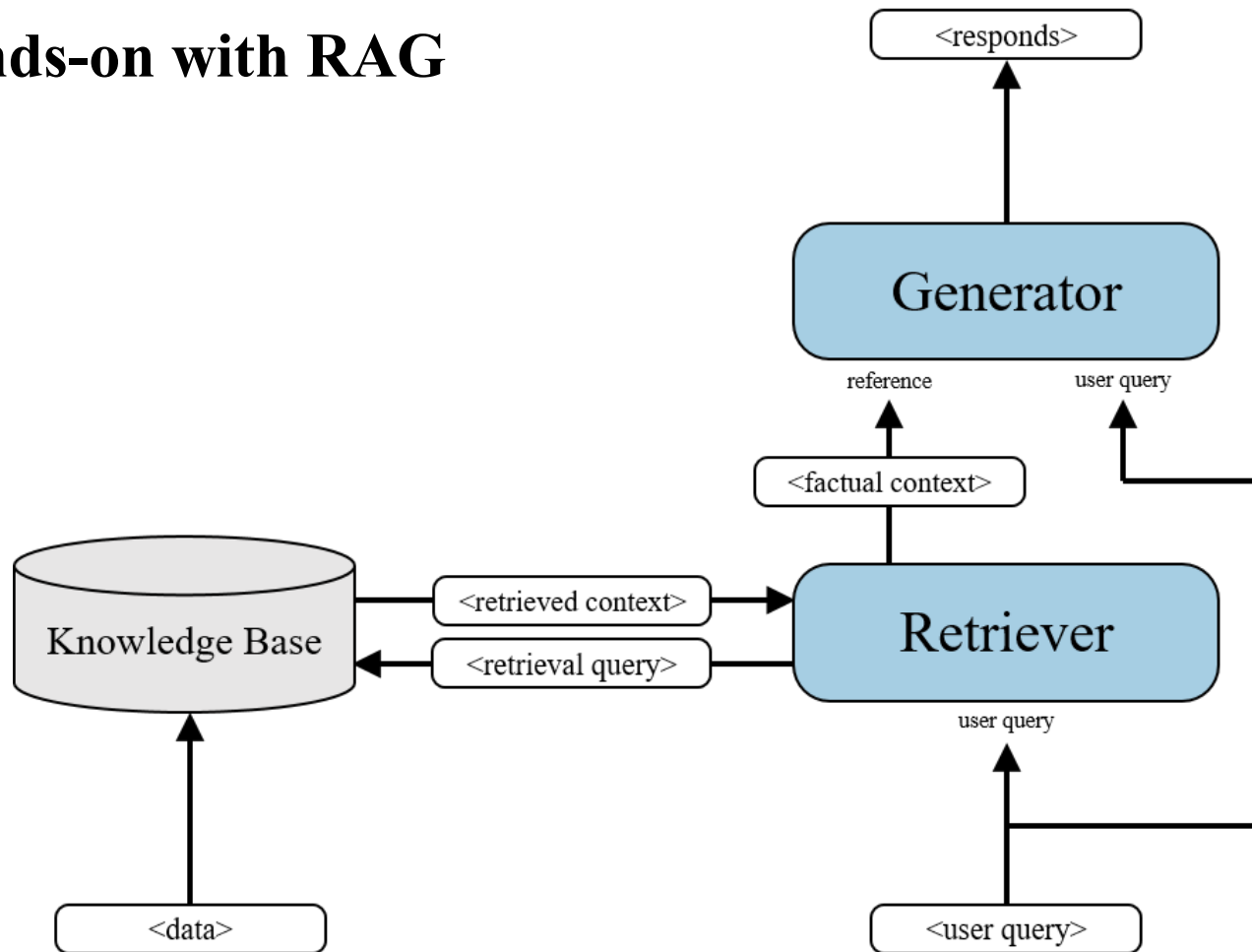
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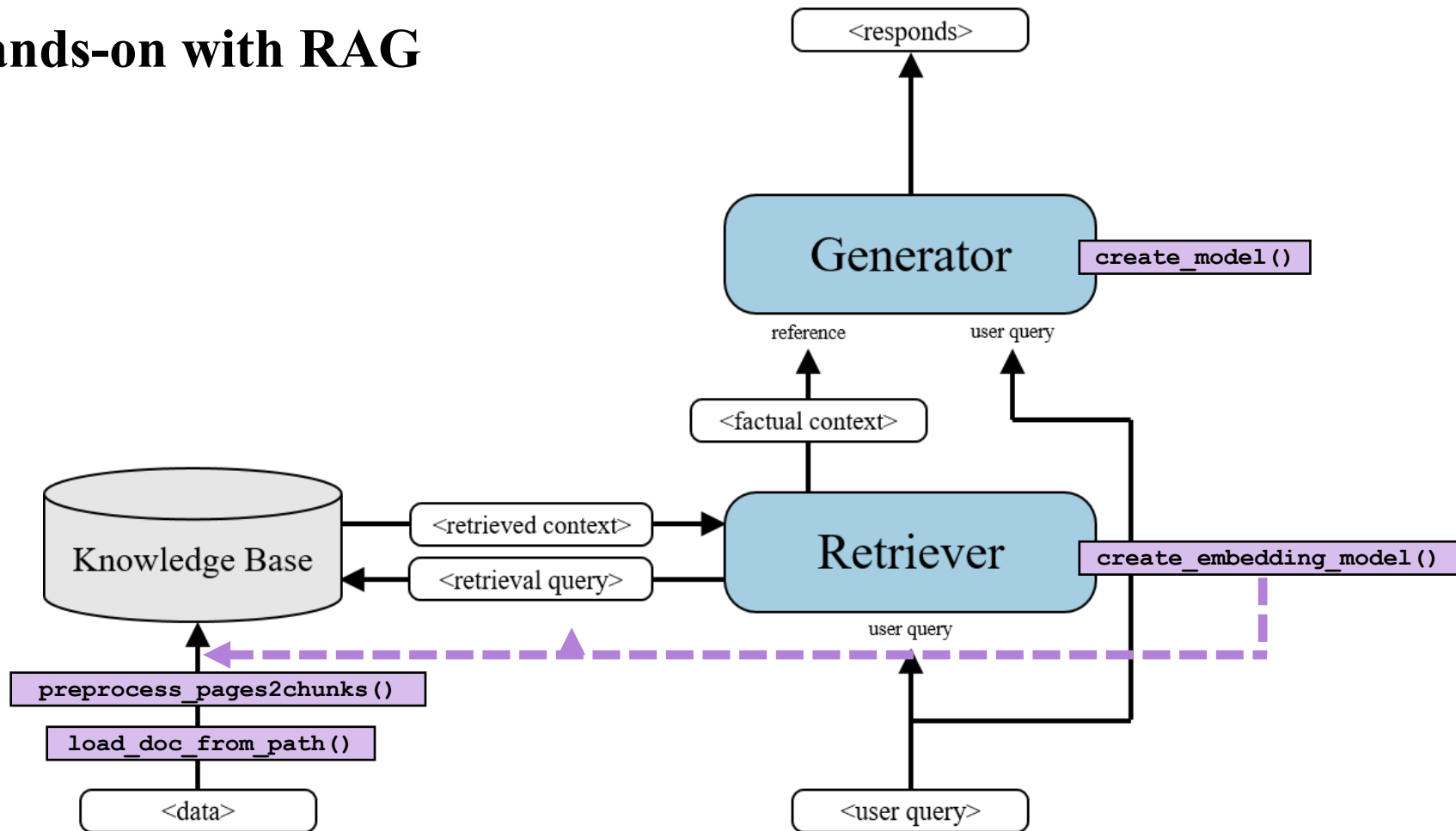
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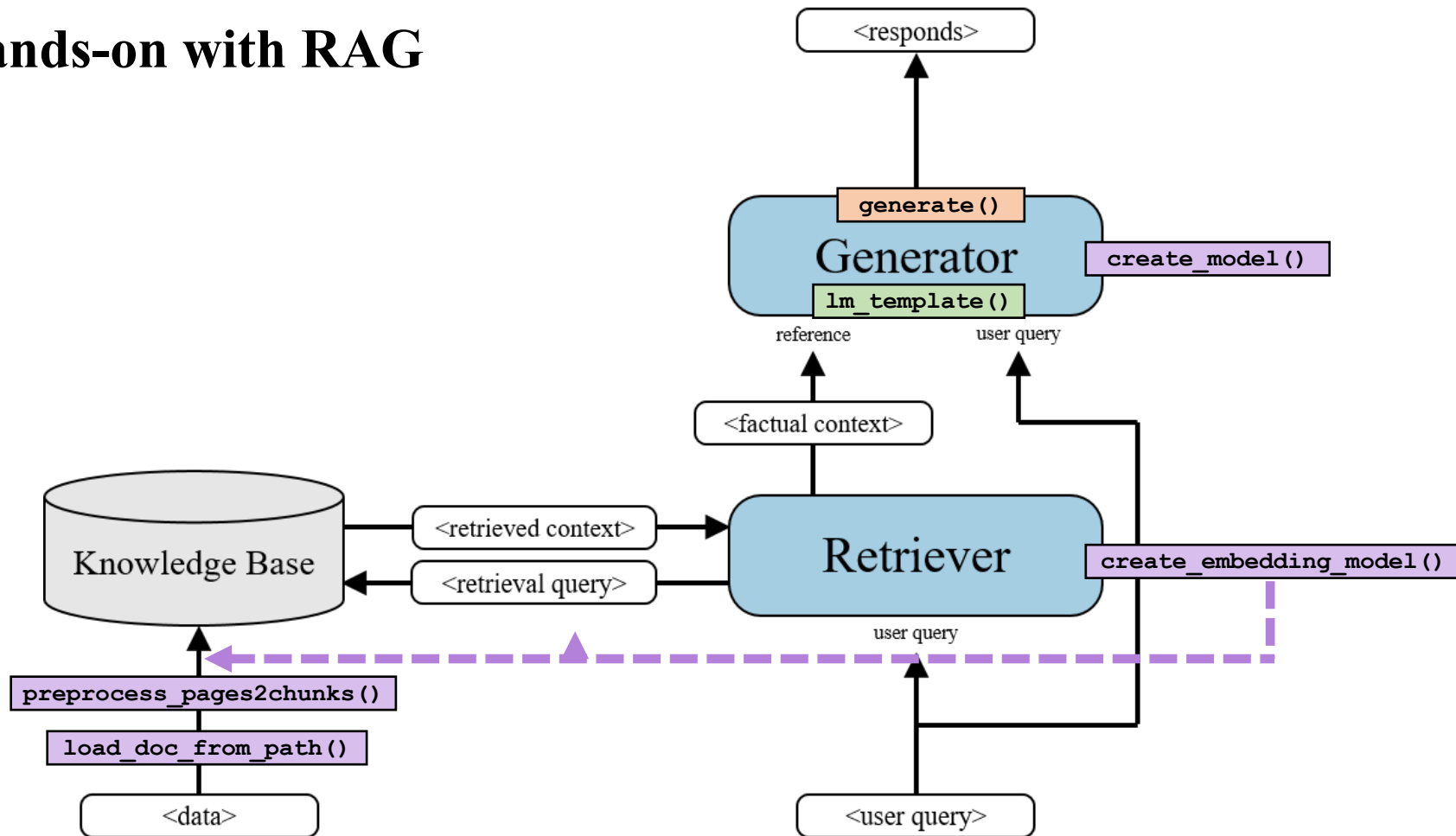
Hands-on with RAG



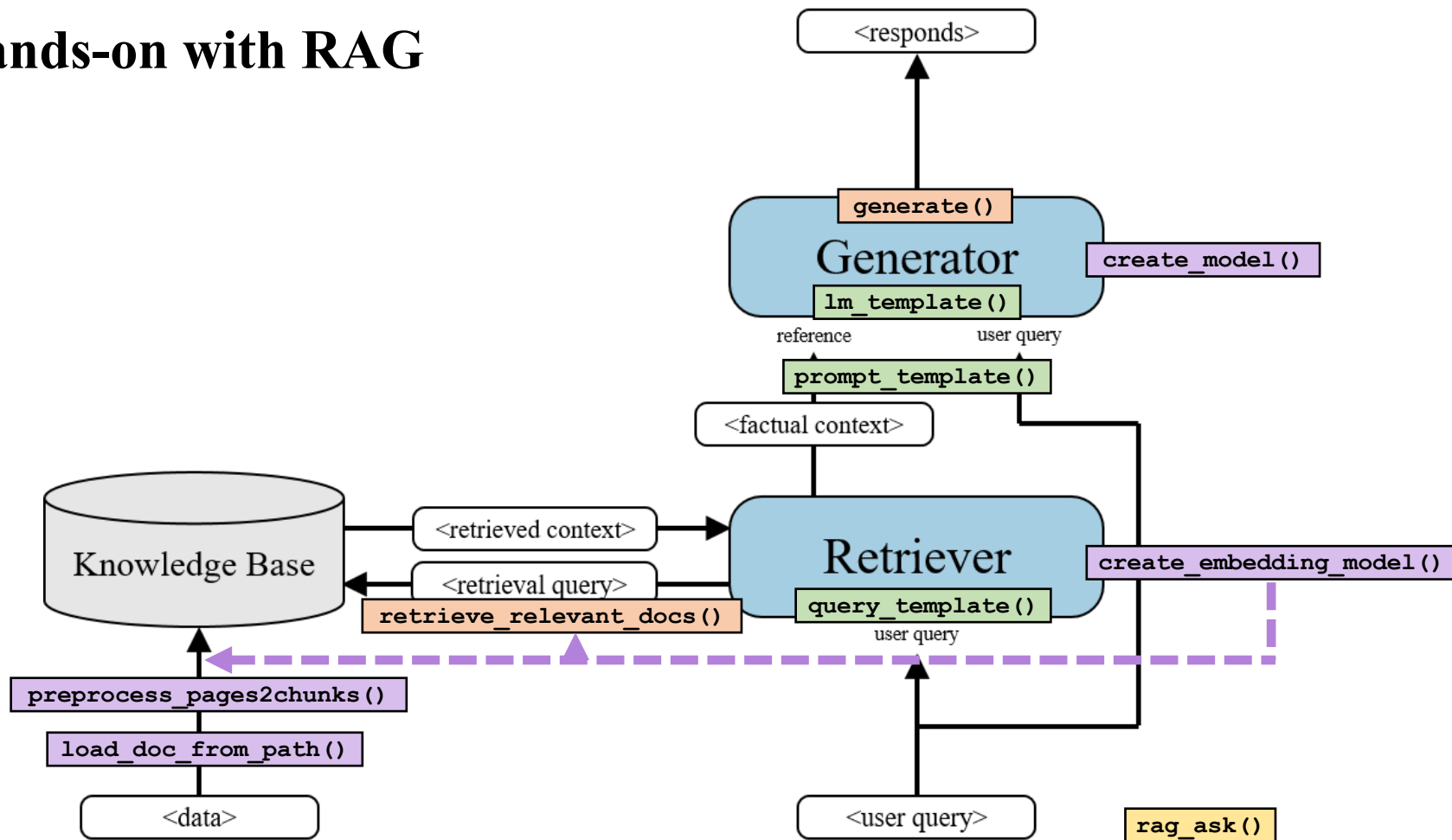
Hands-on with RAG



Hands-on with RAG



Hands-on with RAG



HW3 LLM

Requirements

1. At least compare result with and without RAG

- Must have **evaluation metrics**: BLEU, ROUGE, BERTScore...
- Prepare your own dataset

2. Bonus: compare different models

3. Bonus: try different RAG pipeline

Submit and Grading

1. Jupyter notebook / code

2. Report

- Describe your dataset - 20% ([notebook example](#))
- Result without RAG - 20%
- Result with RAG - 20%
- Compare the result (w/o RAG) - 20%
- Bonus: Different model comparison - 10%
- Bonus: Describe your own RAG pipeline / Compare with sample notebook - 10%

Access model weight

Hugging Face Search models, datasets, users...

Models Datasets Spaces Community Docs Pricing Log In Sign Up

google/gemma-3-4b-it like 905 Follow Google 33.2k

Image-Text-to-Text Transformers Safetensors gemma3 conversational text-generation-inference arxiv:28 papers License: gemma

Model card Files and versions Community 74

Access Gemma on Hugging Face

This repository is publicly accessible, but you have to accept the conditions to access its files and content.

To access Gemma on Hugging Face, you're required to review and agree to Google's usage license. To do this, please ensure you're logged in to Hugging Face and click below. Requests are processed immediately.

Log in or Sign Up to review the conditions and access this model content.

Gemma 3 model card

Model Page: [Gemma](#)

Resources and Technical Documentation:

- [Gemma 3 Technical Report](#)
- [Responsible Generative AI Toolkit](#)
- [Gemma on Kaggle](#)
- [Gemma on Vertex Model Garden](#)

Downloads last month: 764,902

Safetensors

Model size: 4B params Tensor type: BF16 Chat template: Files Info

Inference Providers

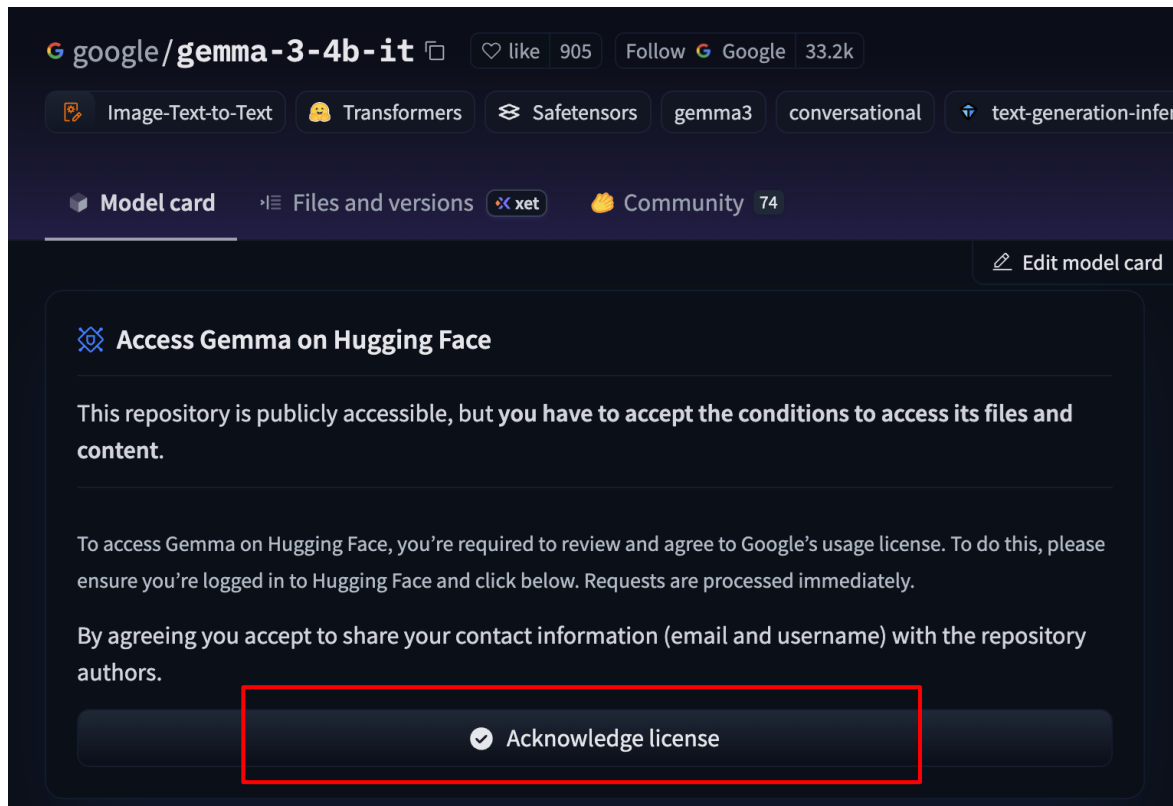
Image-Text-to-Text

This model isn't deployed by any Inference Provider. 61 Ask for provider support

Model tree for google/gemma-3-4b-it

Base model	google/gemma-3-4b-pt
Finetuned (218)	this model
Adapters	86 models
Finetunes	371 models
Merges	2 models
Quantizations	144 models

Access model weight



Access request



Gemma Access Request

✓ Choose Account for Consent



2 Gemma Consent Form



FIRST NAME **

Enter First Name

LAST NAME **

Enter Last Name

VERIFIED EMAIL

a0966096979@gmail.com

Gemma's terms of use were last updated on March 25, 2025. You can view the previous version [here](#).

Gemma Terms of Use

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Section 1

DEFINITIONS

1.1 Definitions

Run on colab

```
!huggingface-cli login --token hf_XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

The screenshot shows the Hugging Face website interface. The top navigation bar includes the Hugging Face logo, a search bar, and links for Models, Datasets, Spaces, Community, Docs, Pricing, and a user profile icon. Below the navigation bar, a banner reads "Hugging Face is way more fun with friends and colleagues! Join an organization". The left sidebar contains a "New" button and a user profile section for "Lunarr1010" with links to Profile, Inbox (0), Settings, Billing, and Get PRO. Below this is the "Organizations" section with a "Create New" button, and a "Resources" section with links to Getting Started, Documentation, Forum, Tasks, and Learn. The main content area is divided into three columns. The first column is titled "Following 0" and lists AI creators to follow: kuri-pl, mlabonne, and Xenova. The second column is titled "Trending" and lists popular models like deepseek-ai/De, PaddlePaddle/P, veo3.1-fast, Qwen/Qwen3-VL, nanonets/Nanon, DeepSite v3, and Sora 2. The third column is a user profile menu for "Lunarr1010" which is highlighted with a red box. This menu includes links to Profile, Notifications, Inbox (0), New Model, New Dataset, New Space, New Collection, Create organization, Usage Quota, Private Storage (0 GB/100 GB), Zero GPU (0/4 min), Inference Usage (\$0.00 / \$0.10), Get Hugging Face PRO, Settings, Access Tokens, Billing, Changelog (3 updates), and Sign Out. At the bottom, there are more model cards like Wan2.2 Animate and karpathy/fineweb-edu-100b-shuffle.

Access Tokens

User Access Tokens

+ Create new token

Access tokens authenticate your identity to the Hugging Face Hub and allow applications to perform actions based on token permissions.  **Do not share your Access Tokens with anyone**; we regularly check for leaked Access Tokens and remove them immediately.

You have no Access Token



Create new Access Token

Token type

Fine-grained

Read

Write

ⓘ This cannot be changed after token creation.

Token name

This token has read-only access to all your and your orgs resources and can make calls to Inference Providers on your behalf. It can also be used to open pull requests and comment on discussions.

Create token

Referencies

[Sentence transformers](#)

[LLM Models](#)